

ES827 – ROBÓTICA INDUSTRIAL

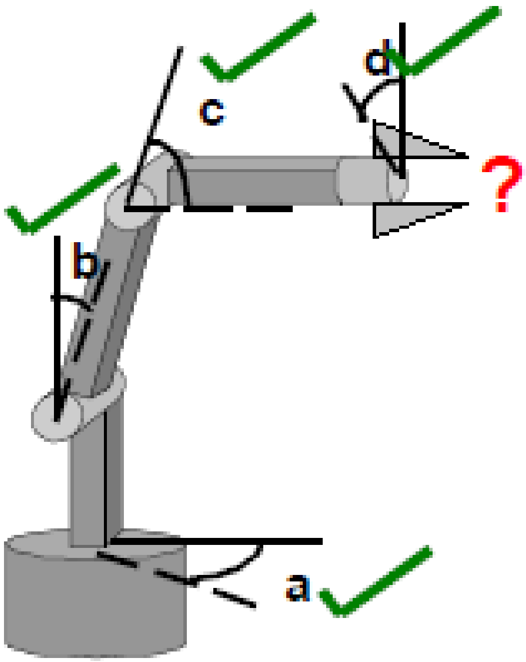
Profª Ludmila Corrêa de Alkmin e Silva (ludmila@fem.unicamp.br)

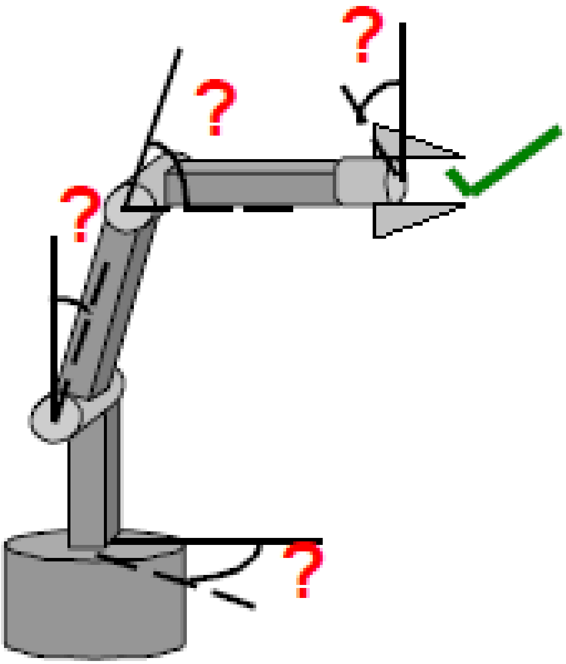
PED : Jonas Cadorini Neto (j299231@dac.unicamp.br)

CINEMÁTICA

Direta

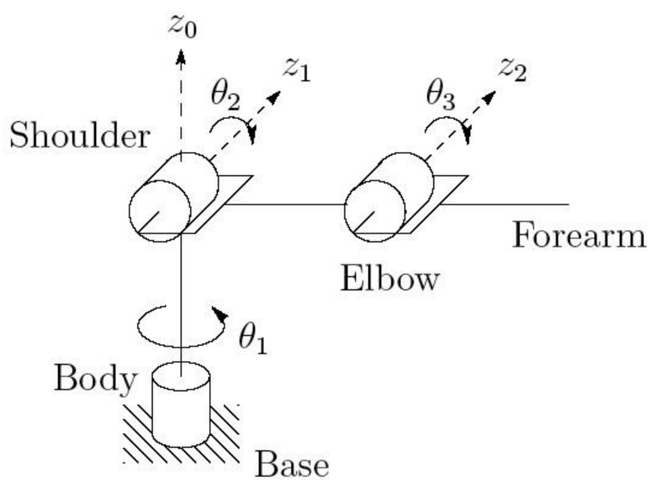
Inversa





ROBÔ

- Um robô manipulador é constituído de elos (links) conectados por juntas.
- Cada junta representa a conexão entre dois elos (links), considerados corpos rígidos.
- O número de juntas define o grau de liberdade do robô.
- As juntas podem ser: prismáticas ou de revolução.
- O punho é a junta que une o último elo ao efetuador final.



DENAVIT-HARTENBERG (DH)

- Representar cada transformação homogênea individual como o produto de quatro transformações básicas:

$$A_i = \mathbf{Rot}_{z,\theta_i}\mathbf{Trans}_{z,d_i}\mathbf{Trans}_{x,a_i}\mathbf{Rot}_{x,\alpha_i}$$

$$= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

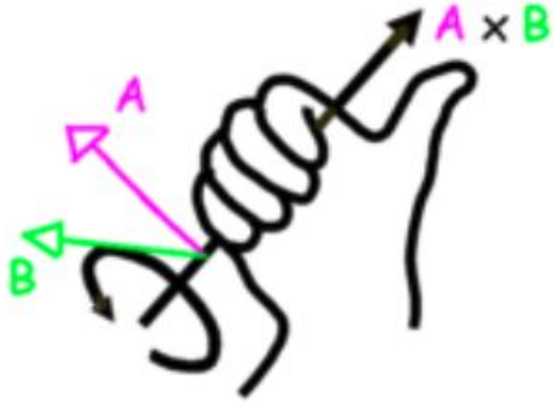
$$= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

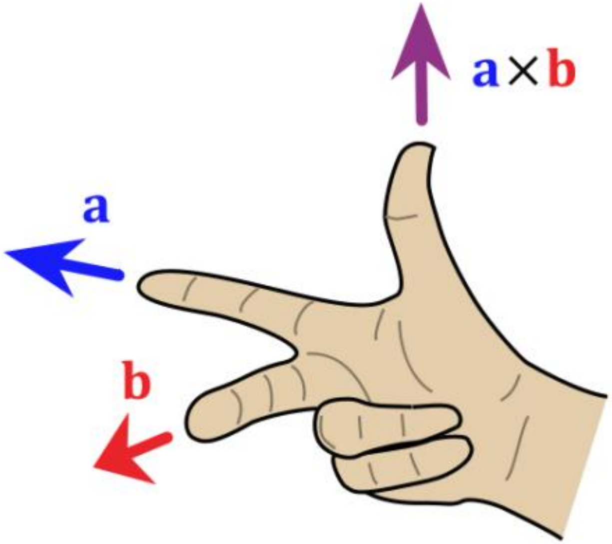
LIMITAÇÕES DENAVIT–HARTENBERG (DH)

- Somente para cadeias cinemáticas abertas de corpos rígidos;
- Cada junta apresenta um único grau de liberdade de translação ou rotação;
- Os diferentes elos do robô são numerados em ordem crescente
- (Base = Elo 0 e Ferramenta = Elo N);
- Convenção rigorosa para a definição dos Sistemas de Coordenadas adotados, como também para as coordenadas de posição e orientação.

REGRAS DA MÃO DIREITA

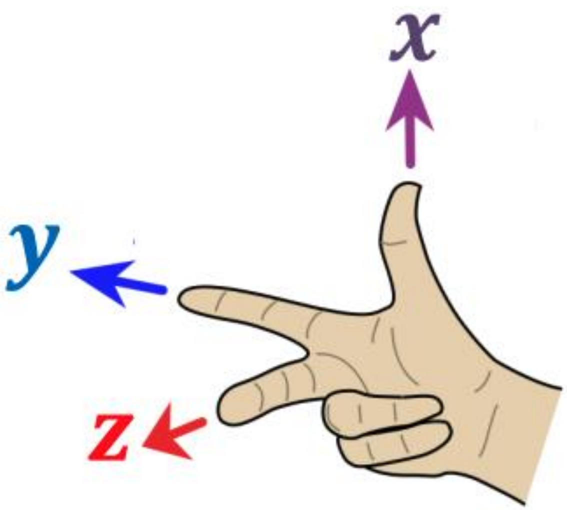
- Produto vetorial

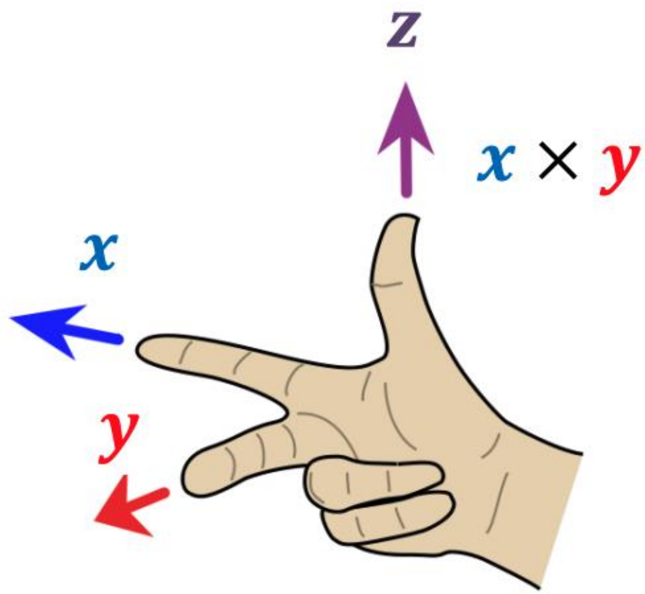




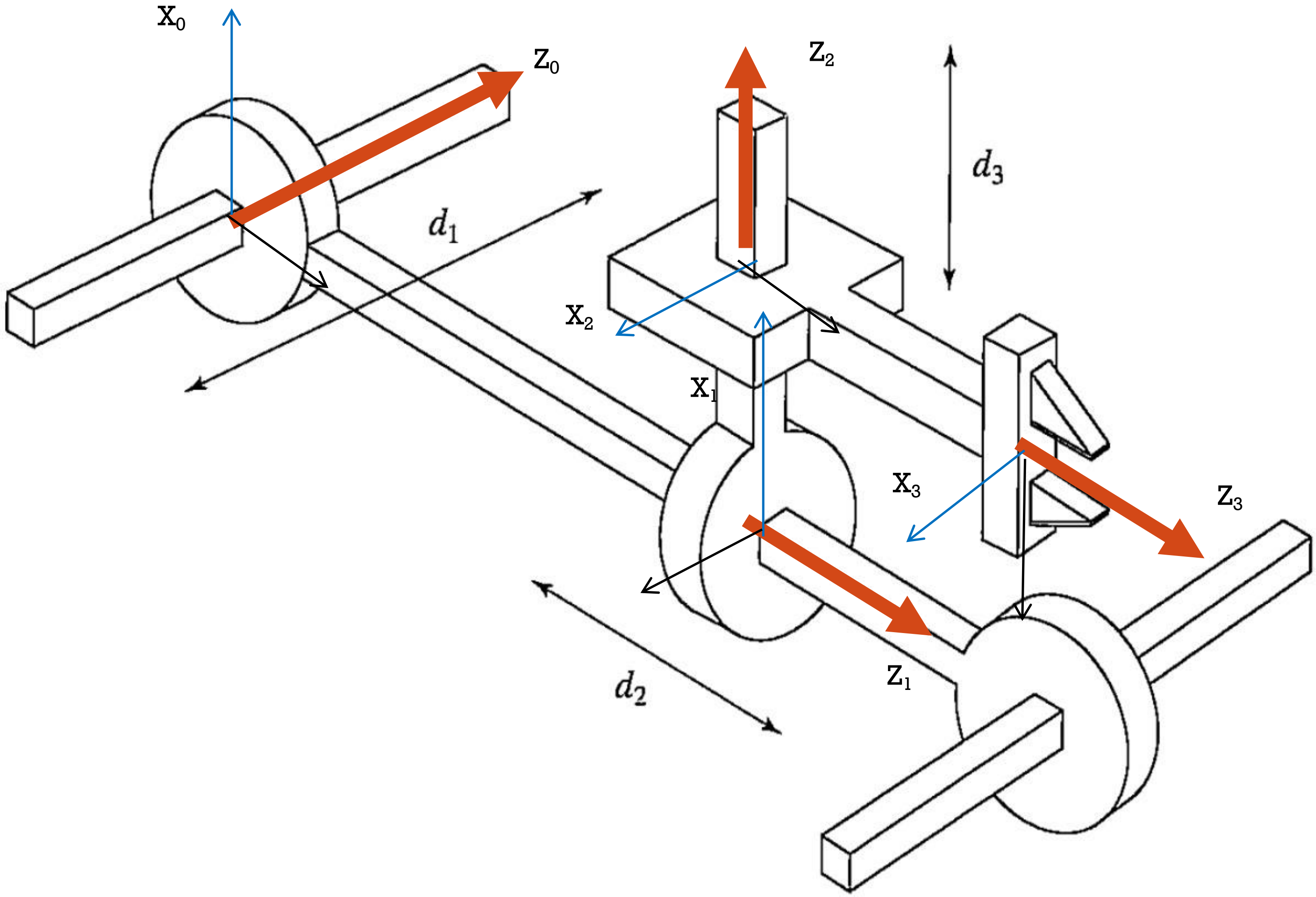
REGRAS DA MÃO DIREITA

- Referências

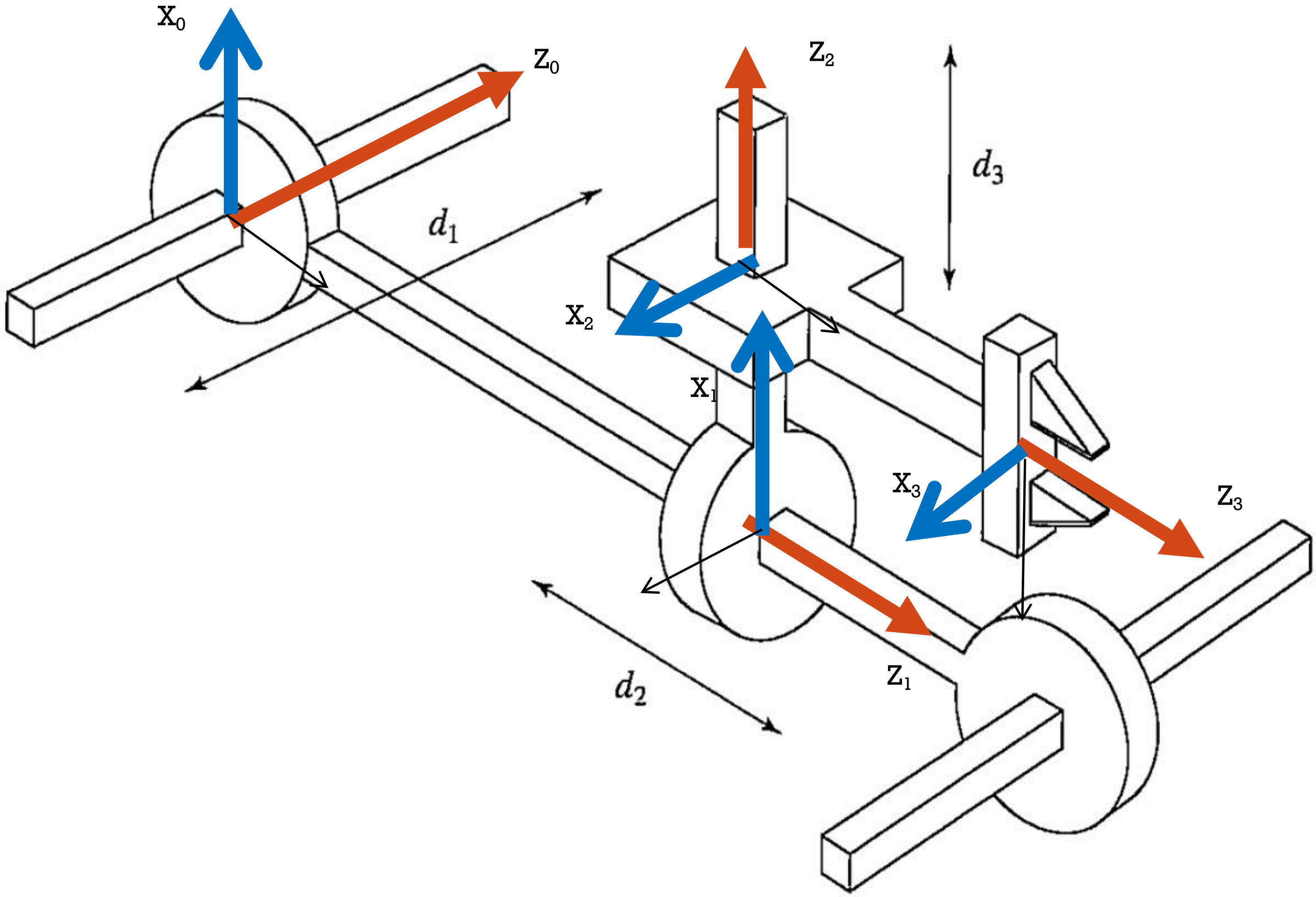




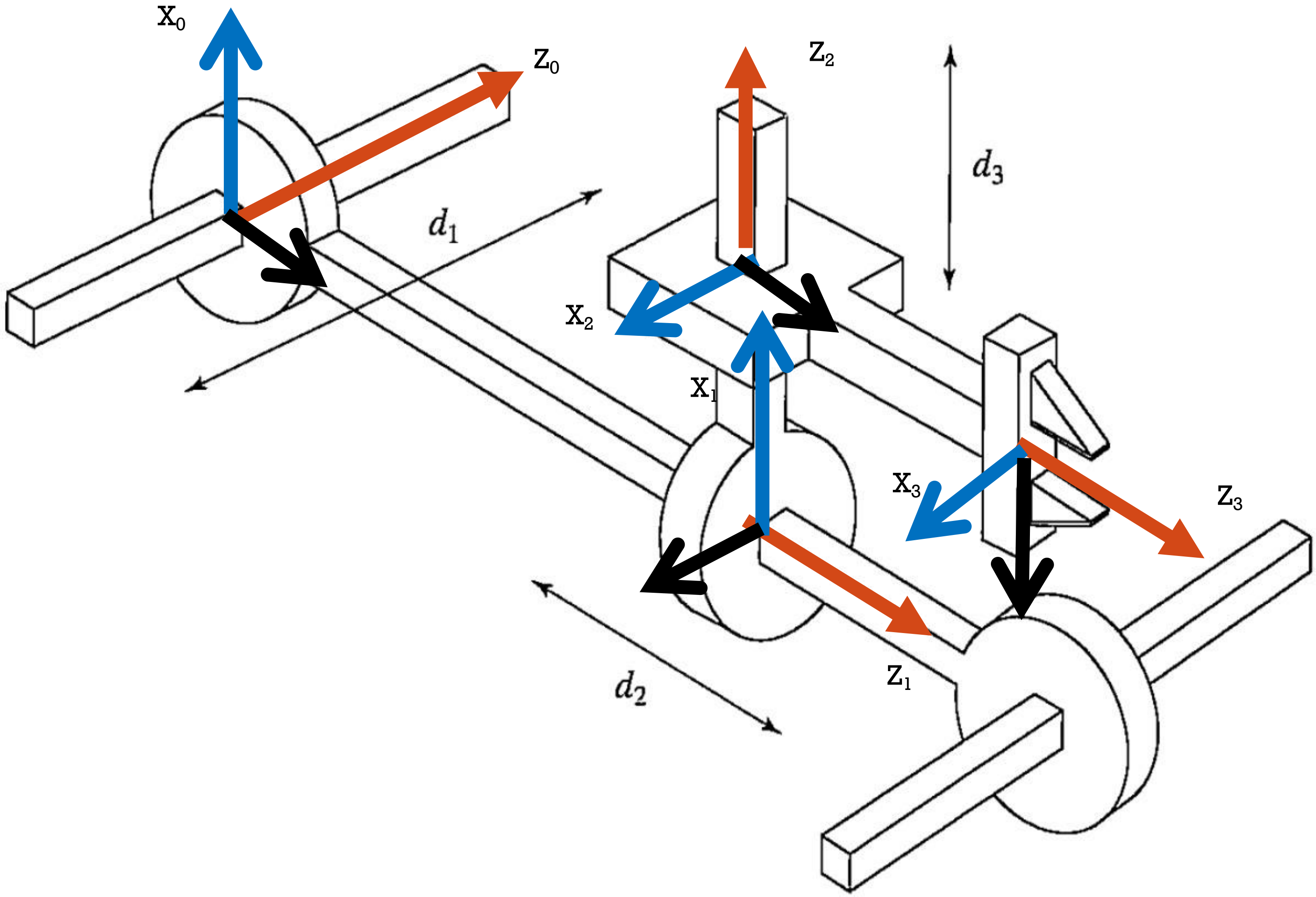
Exemplo 3: Manipulador Cartesiano PPP



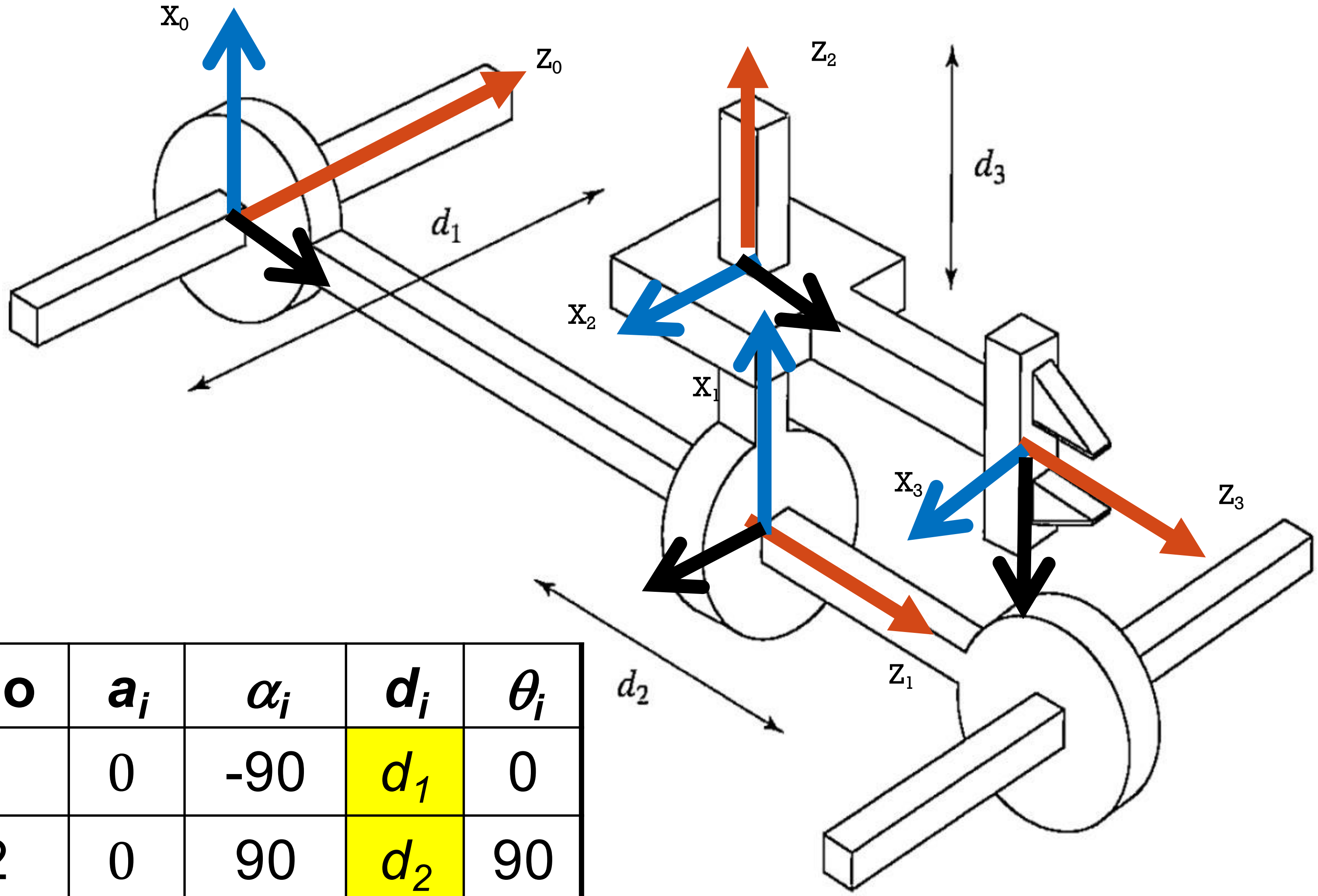
Exemplo 3: Manipulador Cartesiano PPP



Exemplo 3: Manipulador Cartesiano PPP

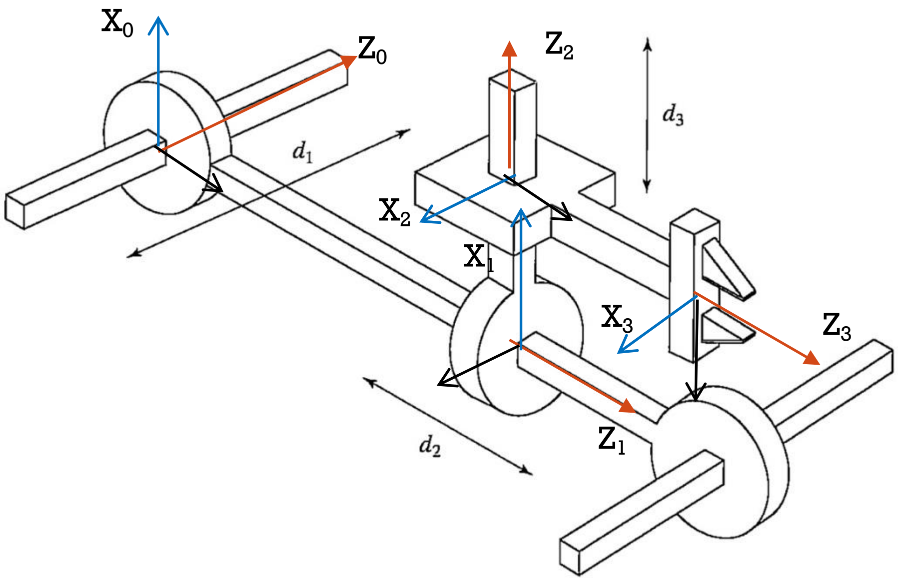


Exemplo 3: Manipulador Cartesiano PPP



Elo	a_i	α_i	d_i	θ_i
1	0	-90	d_1	0
2	0	90	d_2	90
3	0	-90	d_3	0

Exemplo 3: Manipulador Cartesiano PPP



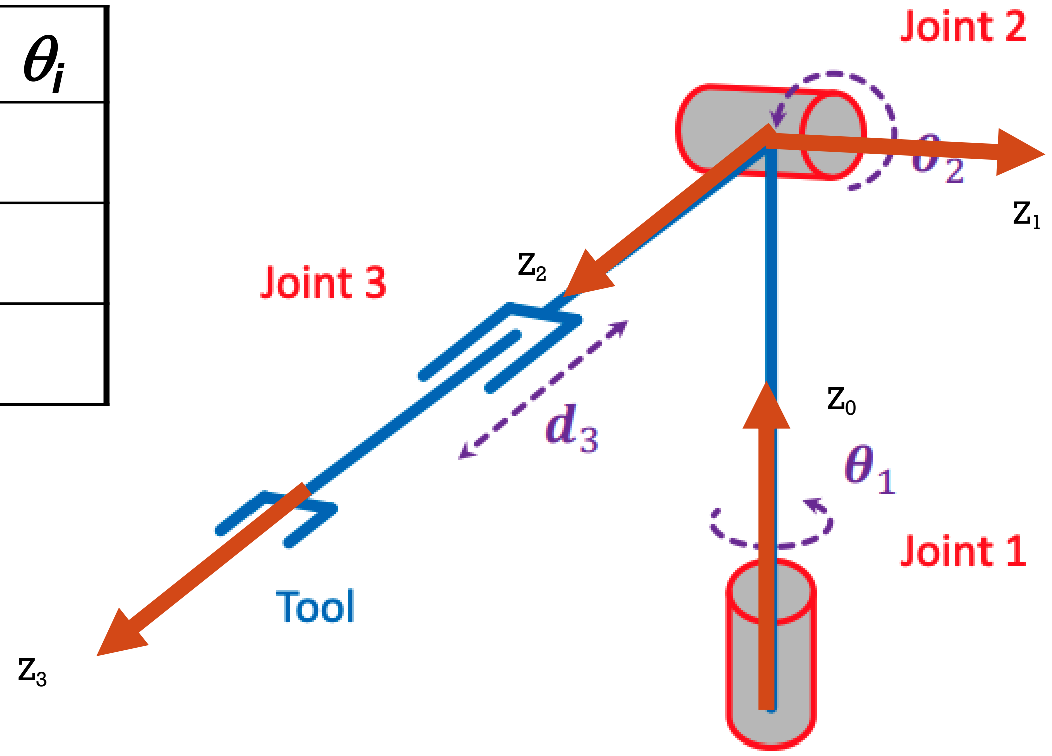
Elo	a_i	α_i	d_i	θ_i
1	0	-90	d_1	0
2	0	90	d_2	90
3	0	-90	d_3	0

$$A_1^0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_2^1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_3^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2^0 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & d_2 \\ -1 & 0 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad A_3^0 = \begin{bmatrix} 0 & -1 & 0 & d_3 \\ 0 & 0 & 1 & d_2 \\ -1 & 0 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

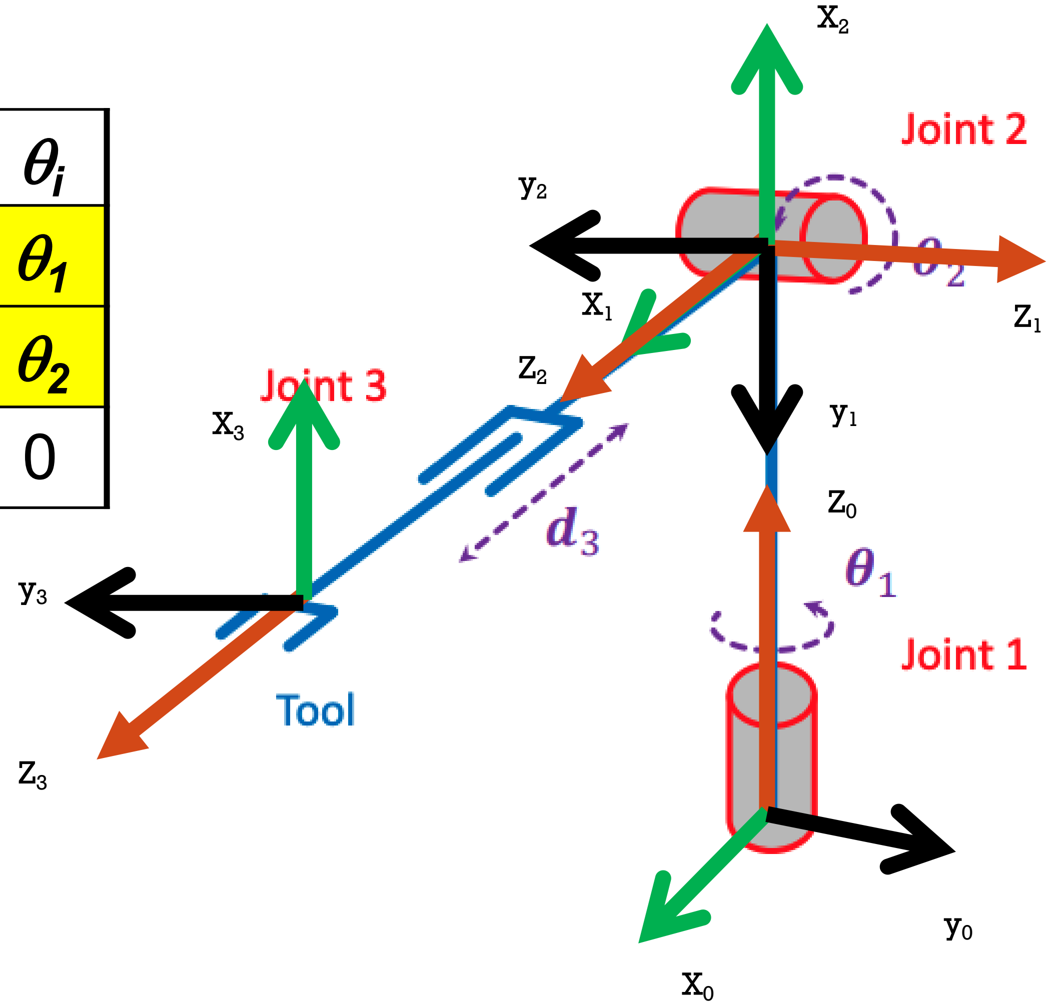
Exemplo 4: Manipulador RRP

Elo	a_i	α_i	d_i	θ_i
1				
2				
3				



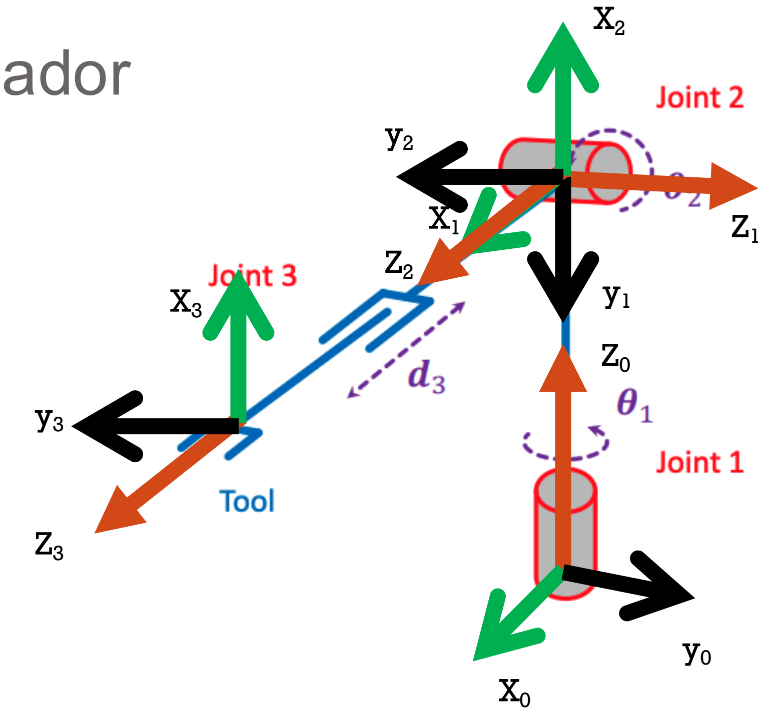
Exemplo 4: Manipulador RRP

Elo	a_i	α_i	d_i	θ_i
1	0	-90	d_1	θ_1
2	0	-90	0	θ_2
3	0	0	d_3	0



Exemplo 4: Manipulador

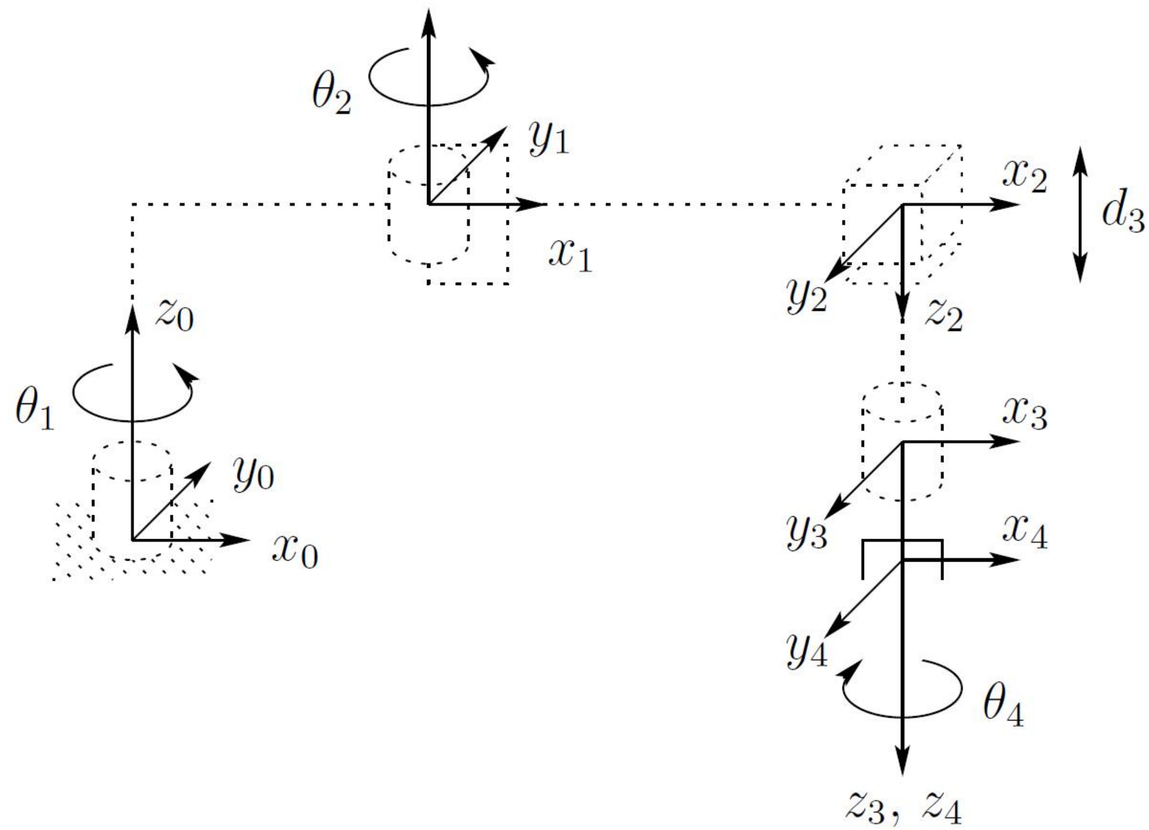
Elo	a_i	α_i	d_i	θ_i
1	0	-90	d_1	θ_1
2	0	-90	0	θ_2
3	0	0	d_3	0



$$A_1^0 = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_2^1 = \begin{bmatrix} c_2 & 0 & -s_2 & 0 \\ s_2 & 0 & c_2 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_3^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

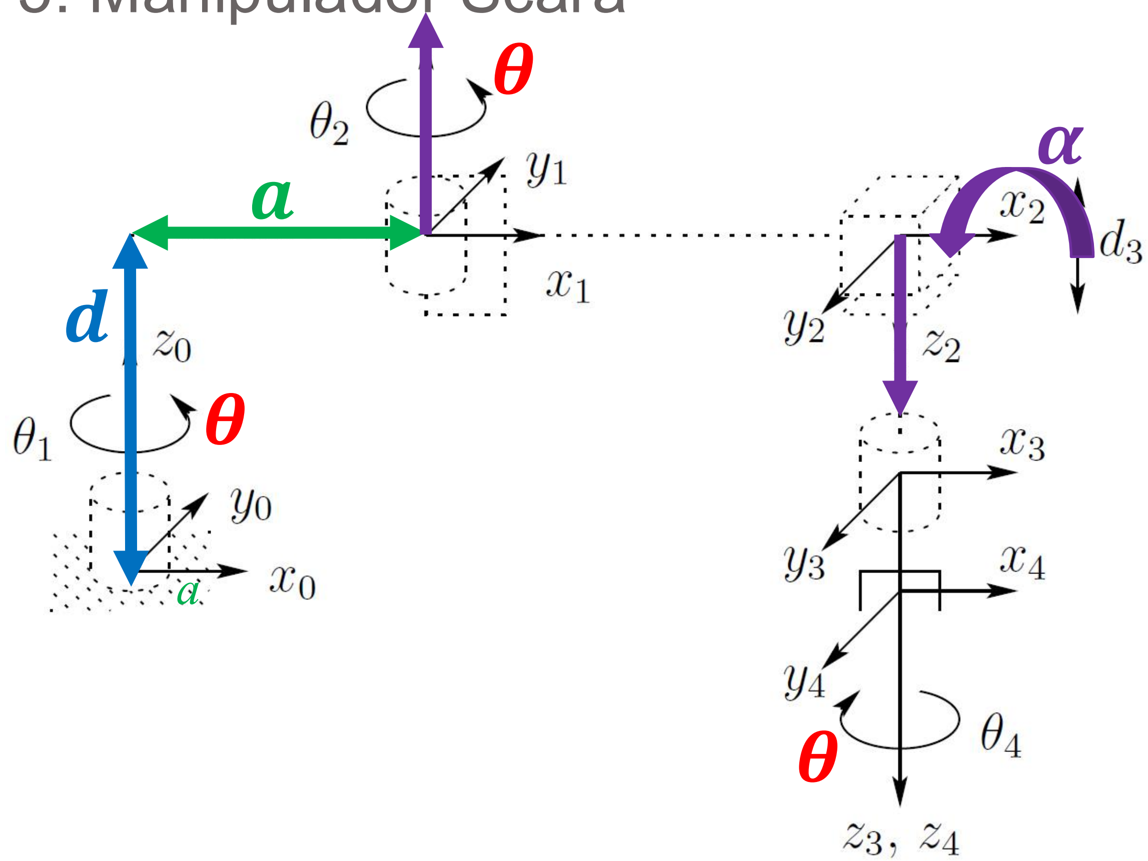
$$A_2^0 = \begin{bmatrix} c_1c_2 & s_1 & -c_1s_2 & 0 \\ s_1c_2 & -c_1 & s_1s_2 & 0 \\ -s_2 & 0 & -c_2 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_3^0 = \begin{bmatrix} c_1c_2 & s_1 & -c_1s_2 & -d_3c_1s_2 \\ s_1c_2 & -c_1 & s_1s_2 & -d_3s_1s_2 \\ -s_2 & 0 & -c_2 & d_1 - d_3c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Exemplo 5: Manipulador Scara



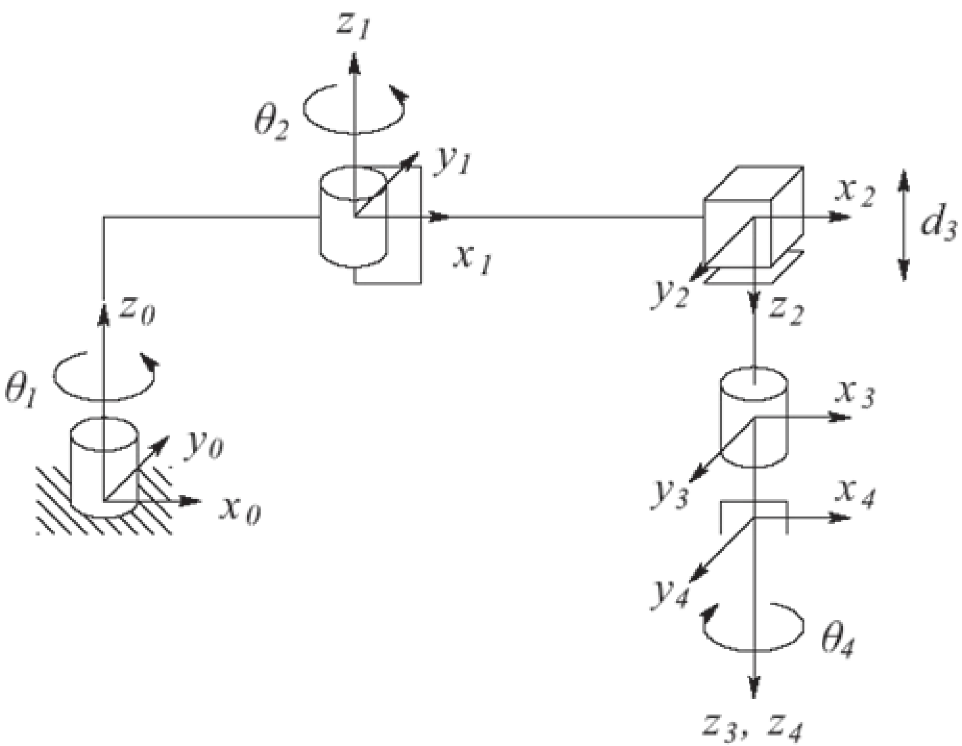
[redacted]

Exemplo 5: Manipulador Scara



- a: Distância (no eixo x_1) entre z_0 e z_1
- α: Ângulo de rotação (no eixo x_1) entre z_0 e z_1
- d: Distância (no eixo z_0) entre O_0 e a interseção entre z_0 e x_1
- θ: Ângulo de rotação (no eixo z_0) entre x_0 e x_1

Exemplo 5: Manipulador Scara



$$A_i = \begin{vmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

link	a_i	α_i	d_i	θ_i
1	a_1	0	d_1	θ_1
2	a_2	180	0	θ_2
3	0	0	d_3	0
4	0	0	d_4	θ_4

Exemplo 5: Manipulador Scara

$$A_1 = \begin{bmatrix} c_1 & -s_1 & 0 & a_1c_1 \\ s_1 & c_1 & 0 & a_1s_1 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, A_2 = \begin{bmatrix} c_2 & s_2 & 0 & a_2c_2 \\ s_2 & -c_2 & 0 & a_2s_2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, A_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}, A_4 = \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

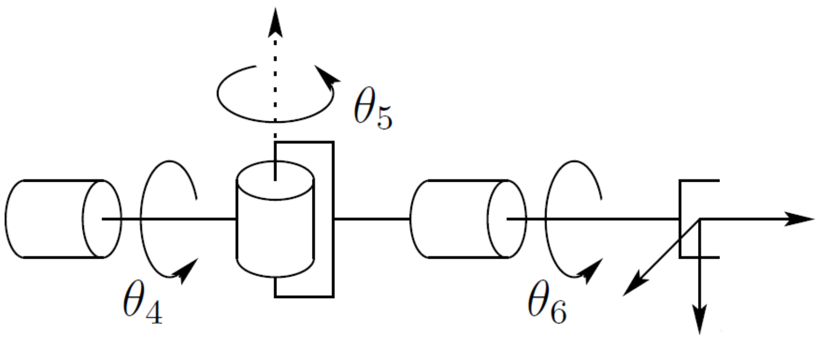
$$T_4^0 = A_1 \cdots A_4 = \begin{bmatrix} c_{12}c_4 + s_{12}s_4 & -c_{12}s_4 + s_{12}c_4 & 0 & a_1c_1 + a_2c_{12} \\ s_{12}c_4 - c_{12}s_4 & -s_{12}s_4 - c_{12}c_4 & 0 & a_1s_1 + a_2s_{12} \\ 0 & 0 & -1 & d_1 - d_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matriz de Rotação

Posição

$$T_4^0 = A_1 \cdots A_4 = \begin{bmatrix} c_{12}c_4 + s_{12}s_4 & -c_{12}s_4 + s_{12}c_4 & 0 & a_1c_1 + a_2c_{12} \\ s_{12}c_4 - c_{12}s_4 & -s_{12}s_4 - c_{12}c_4 & 0 & a_1s_1 + a_2s_{12} \\ 0 & 0 & -1 & d_1 - d_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Exemplo 6: Garra Esférica



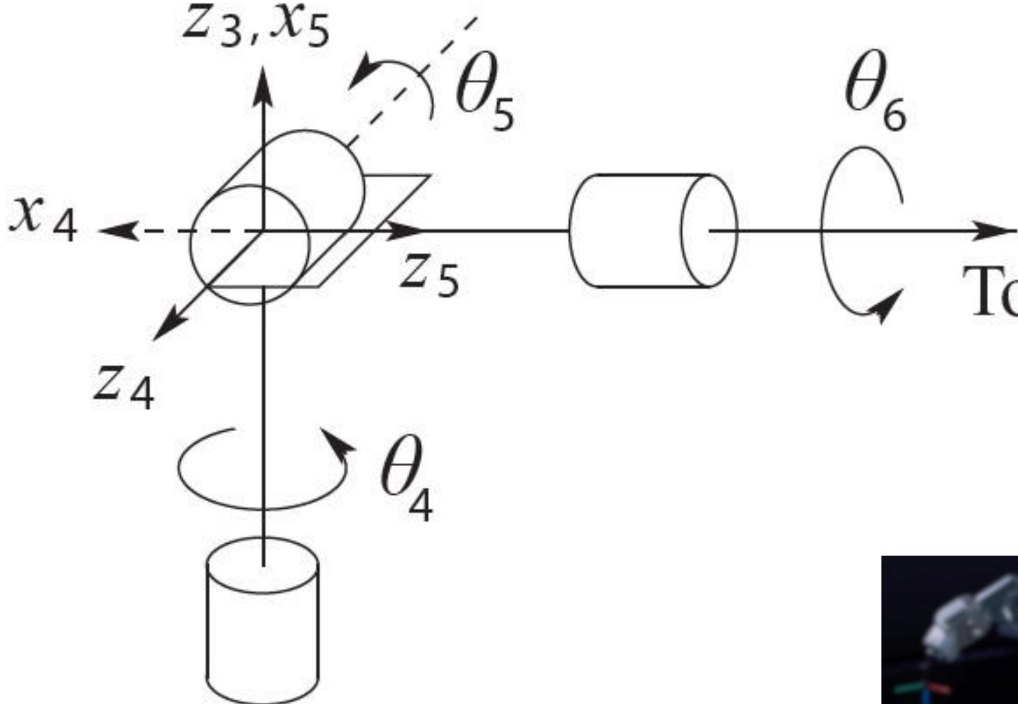
link	a_i	α_i	d_i	θ_i
4	0	-90	0	θ_4
5	0	90	0	θ_5
6	0	0	d_6	θ_6

$$A_4 = \begin{bmatrix} c_4 & 0 & -s_4 & 0 \\ s_4 & 0 & c_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, A_5 = \begin{bmatrix} c_5 & 0 & -s_5 & 0 \\ s_5 & 0 & c_5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, A_6 = \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ s_6 & c_6 & 0 & 0 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

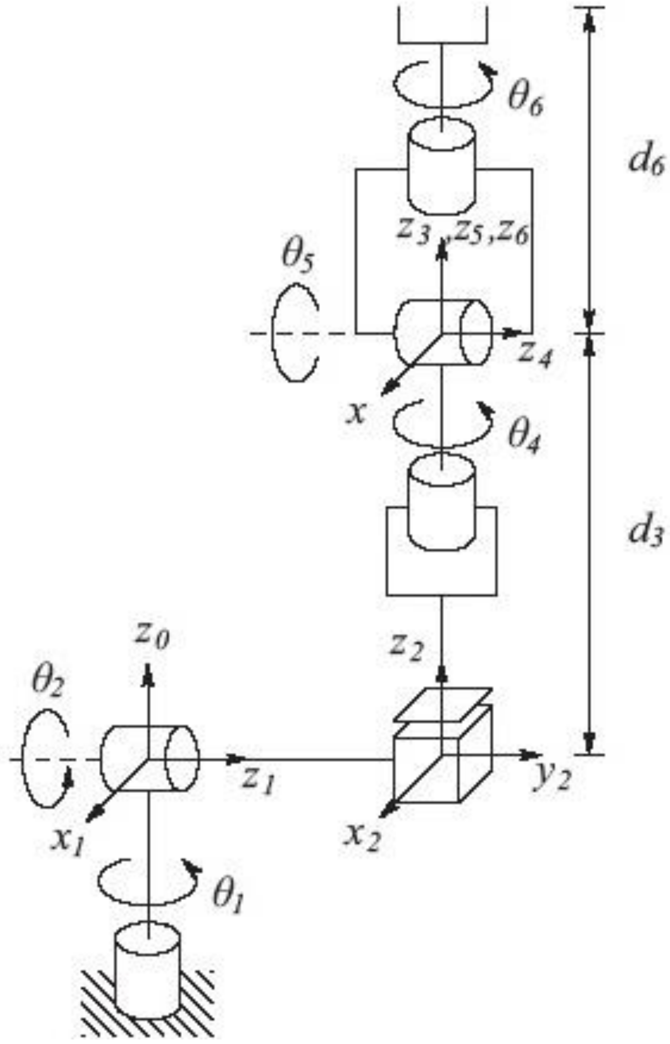
$$T_6^3 = A_4 A_5 A_6 = \begin{bmatrix} c_4c_5c_6 - s_4s_6 & -c_4c_5s_6 - s_4c_6 & c_4s_5 & c_4s_5d_6 \\ s_4c_5c_6 + c_4s_6 & -s_4c_5s_6 + c_4c_6 & s_4s_5 & s_4s_5d_6 \\ -s_5c_6 & s_5c_6 & c_5 & c_5d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

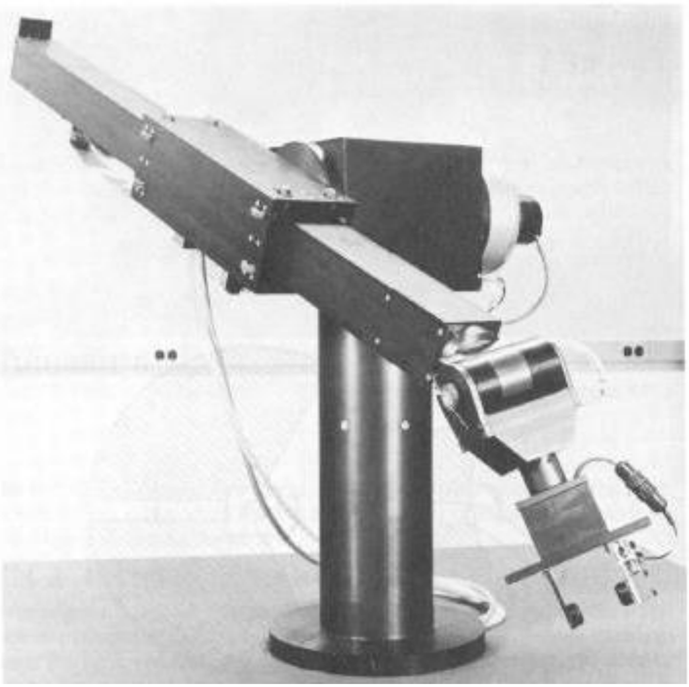
Exemplo 6: Garra Esférica

link	a_i	α_i	d_i	θ_i
4	0	-90	0	θ_4
5	0	90	0	$\theta_5 - 90$
6	0	0	d_6	θ_6

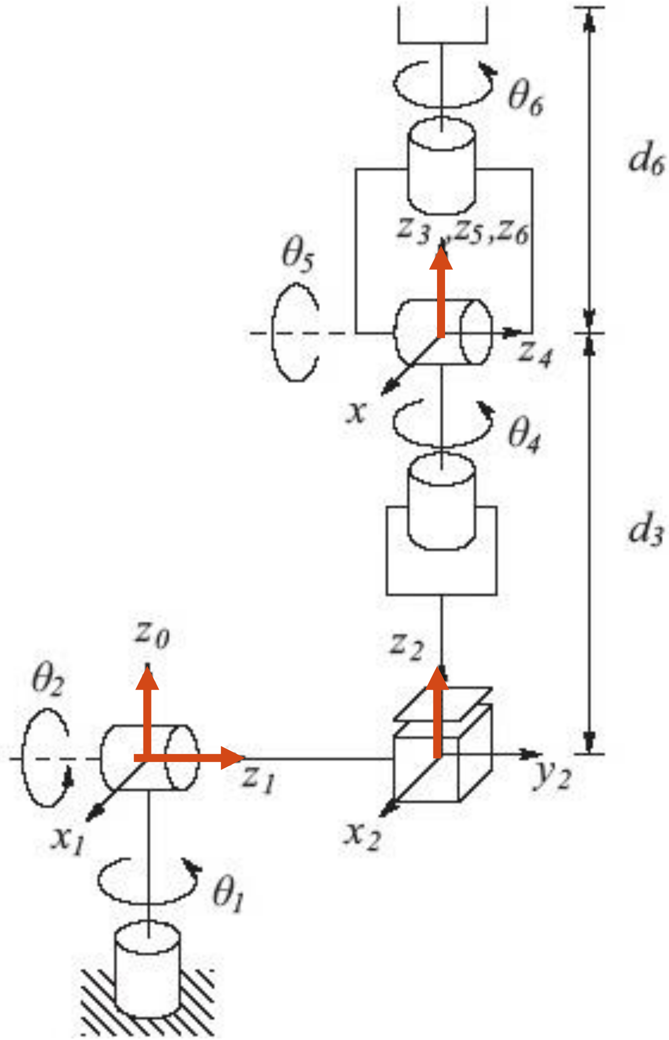


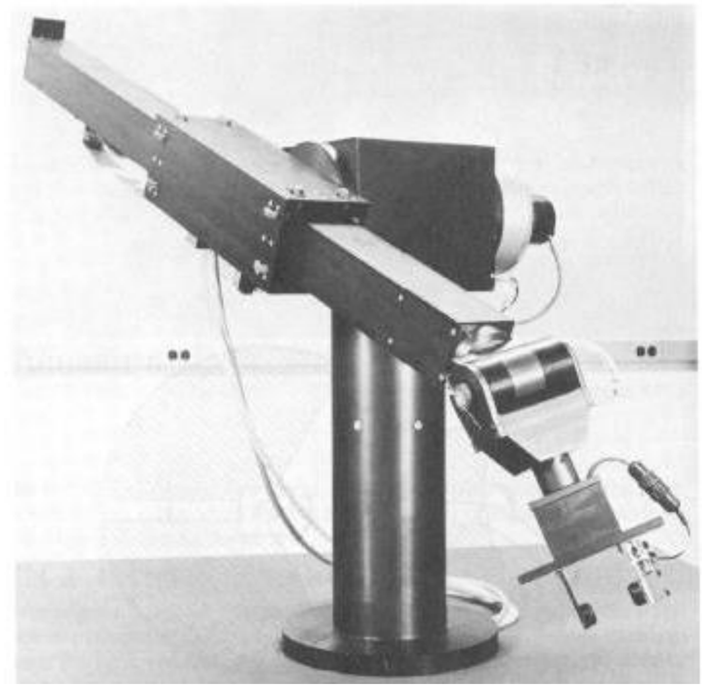
Exemplo 7: Manipulador Stanford



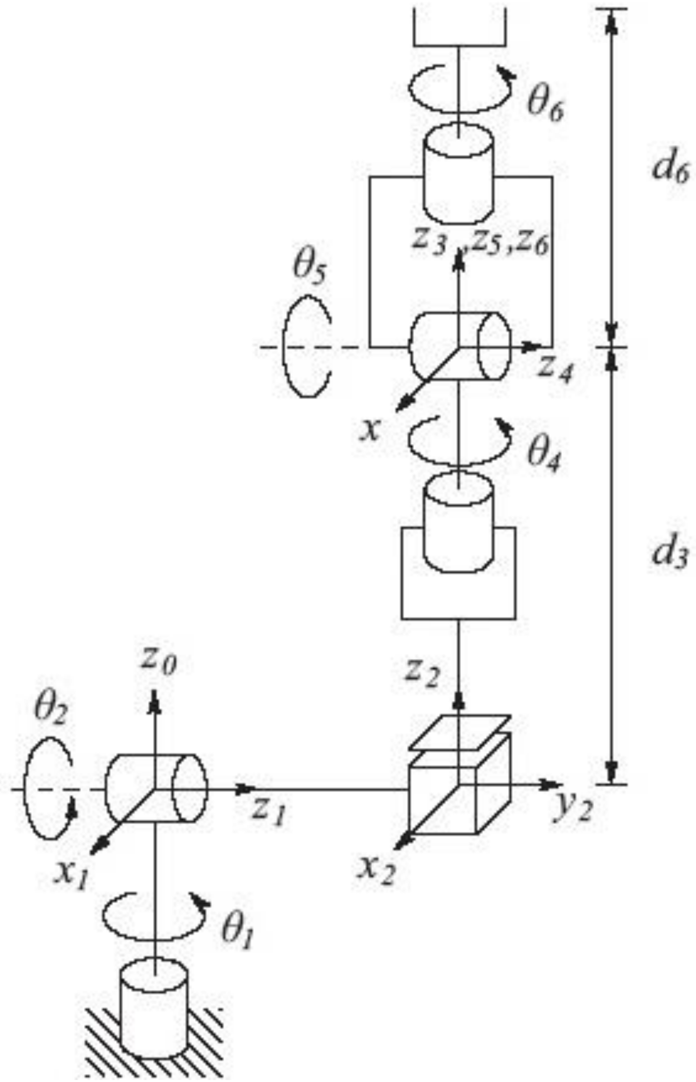


Exemplo 7: Manipulador Stanford





Exemplo 7: Manipulador Stanford



link	a_i	α_i	d_i	θ_i
1	0	-90	0	θ_1
2	0	90	d_2	θ_2
3	0	0	d_3	0
4	0	-90	0	θ_4
5	0	90	0	θ_5
6	0	0	d_6	θ_6

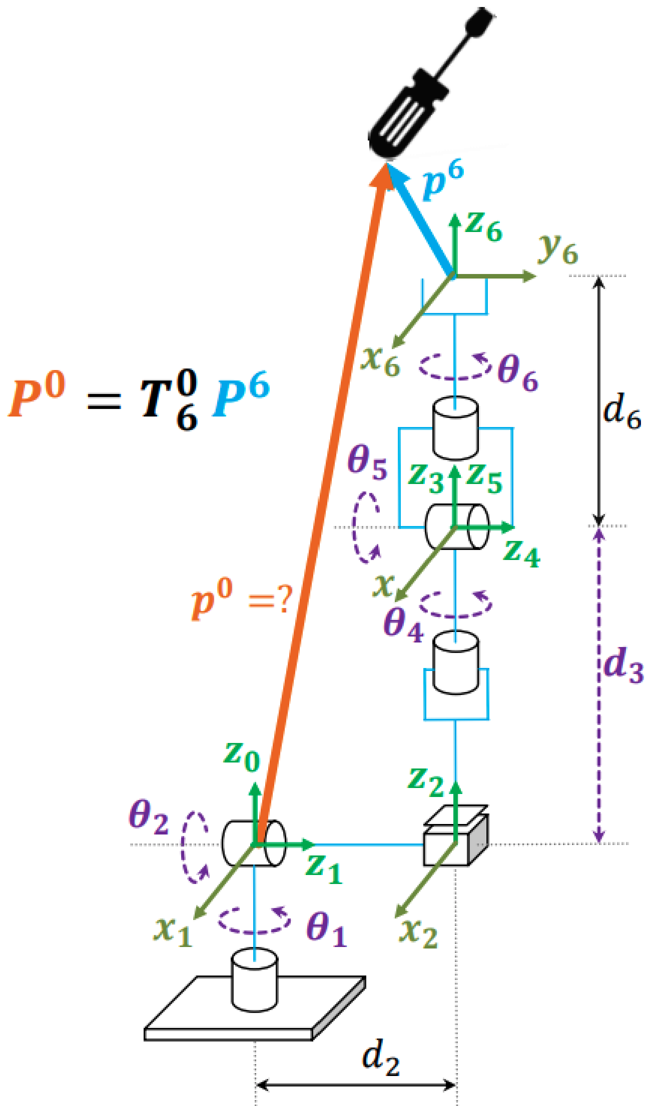
Exemplo 7: Manipulador Stanford

$$T_6^0 = A_1 \cdots A_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & d_x \\ r_{21} & r_{22} & r_{23} & d_y \\ r_{31} & r_{32} & r_{33} & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{cases}
r_{11} = c_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] - d_2(s_4c_5c_6 + c_4s_6) \\
r_{21} = s_1[c_2(c_4c_5c_6 - s_4s_6) - s_2s_5c_6] + c_1(s_4c_5c_6 + c_4s_6) \\
r_{31} = -s_2(c_4c_5c_6 - s_4s_6) - c_2s_5c_6 \\
r_{12} = c_1[-c_2(c_4c_5s_6 + s_4c_6) + s_2s_5s_6] - s_1(-s_4c_5s_6 + c_4c_6) \\
r_{22} = -s_1[-c_2(c_4c_5s_6 - s_4c_6) - s_2s_5s_6] + c_1(-s_4c_5s_6 + c_4s_6) \\
r_{32} = s_2(c_4c_5s_6 + s_4c_6) + c_2s_5s_6 \\
r_{13} = c_1(c_2c_4s_5 + s_2c_5) - s_1s_4s_5 \\
r_{23} = s_1(c_2c_4s_5 + s_2c_5) + c_1s_4s_5 \\
r_{33} = -s_2c_4s_5 + c_2c_5 \\
d_x = c_1s_2d_3 - s_1d_2 + d_6(c_1c_2c_4s_5 + c_1c_5s_2 - s_1s_4s_5) \\
d_y = s_1s_2d_3 + c_1d_2 + d_6(c_1s_4s_5 + c_2c_4s_1s_5 + c_5s_1s_2) \\
d_z = c_2d_3 + d_6(c_2c_5 - c_4s_2s_5)
\end{cases}$$

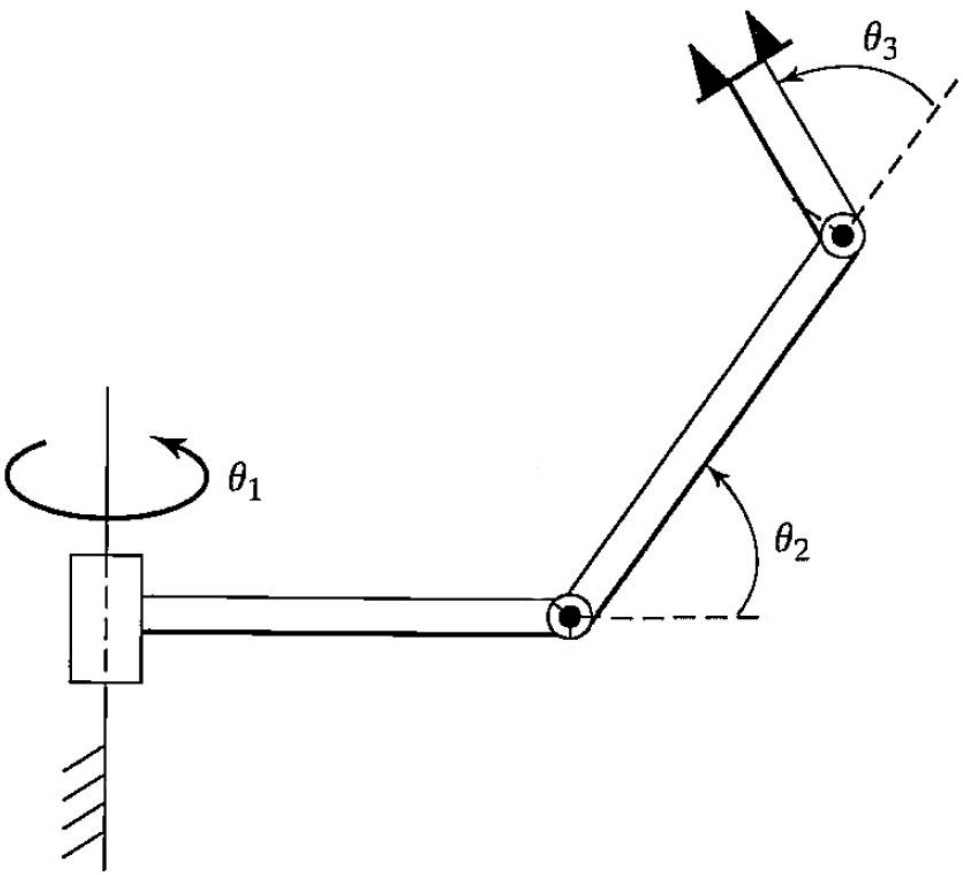
Exemplo 7: Manipulador Stanford

$$T_6^0 = A_1 \cdots A_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & d_x \\ r_{21} & r_{22} & r_{23} & d_y \\ r_{31} & r_{32} & r_{33} & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



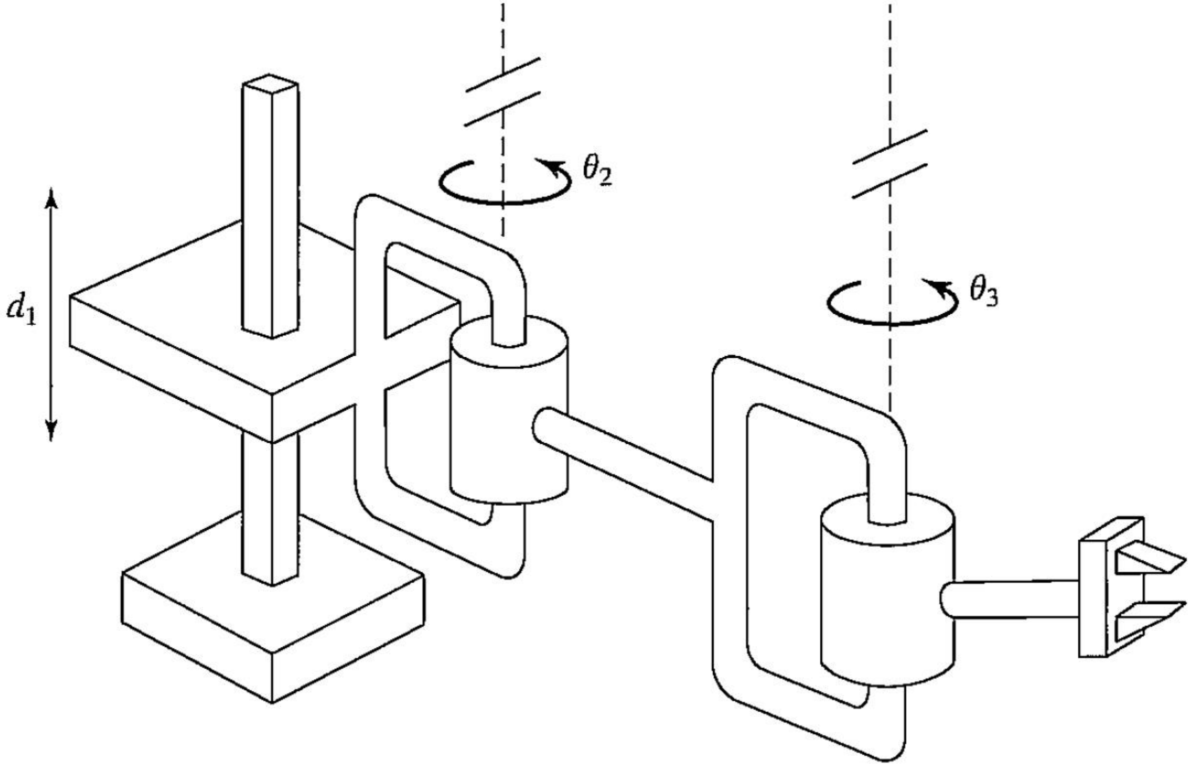
Exemplo 8: 3 GDL RRR

Elo	a_i	α_i	d_i	θ_i
1	a_1	90	d_1	θ_1
2	a_2	0	0	θ_2
3	a_3	0	0	θ_3



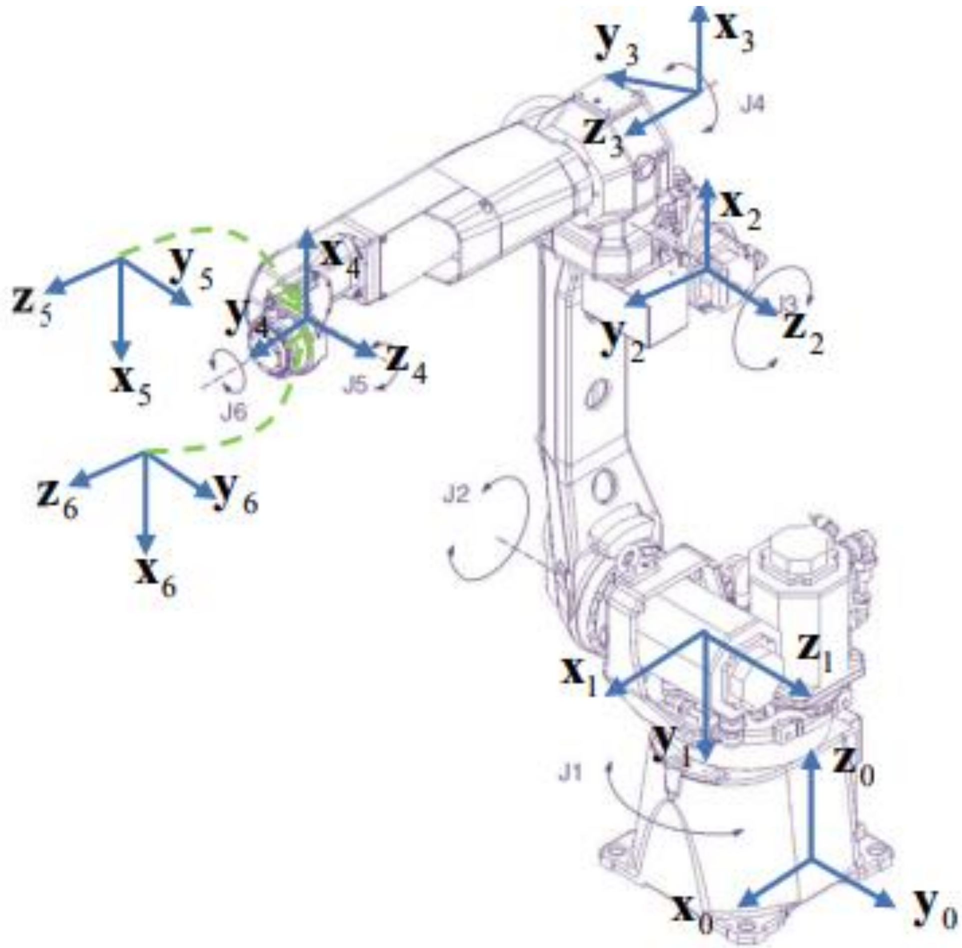
Exemplo 9: 3 GDL PRR

Elo	a_i	α_i	d_i	θ_i
1	a_1	0	d_1	0
2	a_2	0	0	θ_2
3	a_3	0	0	θ_3



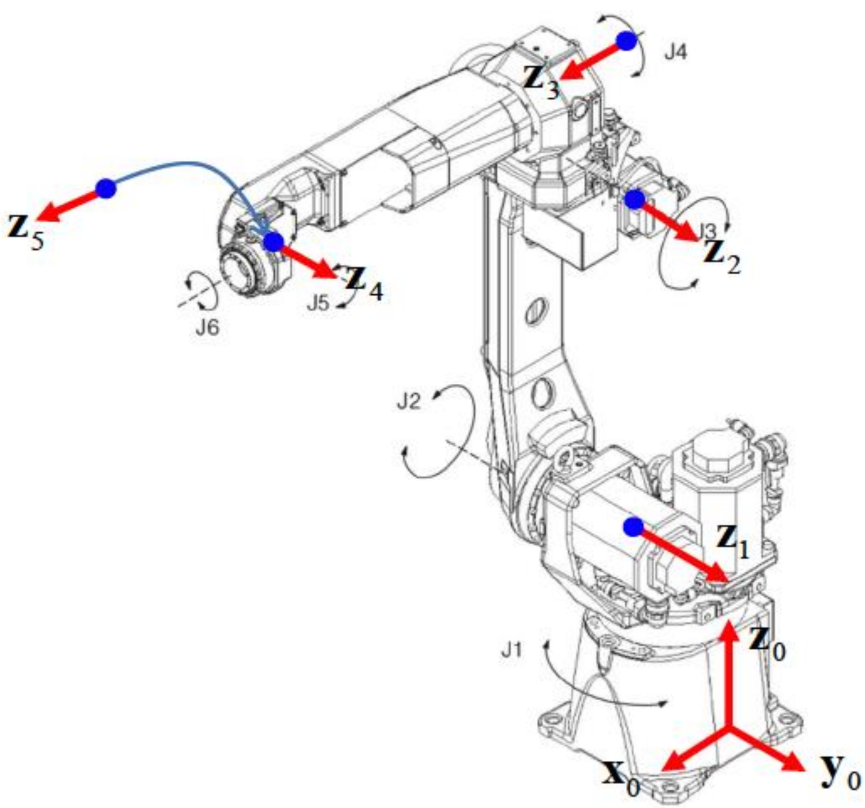
Exemplo 10: Fanuc M-10iA

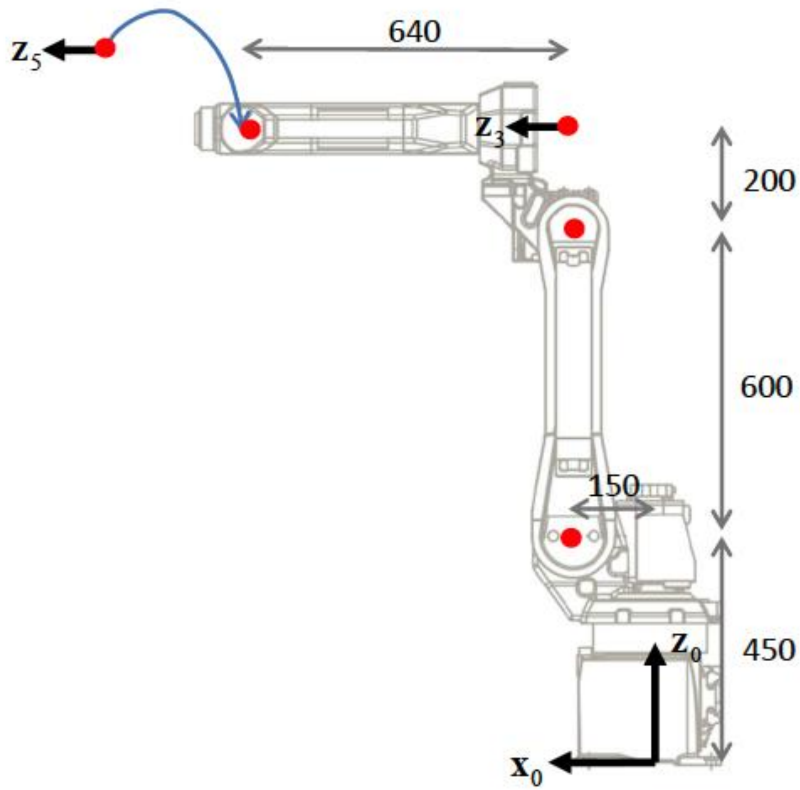
Fanuc M-10iA



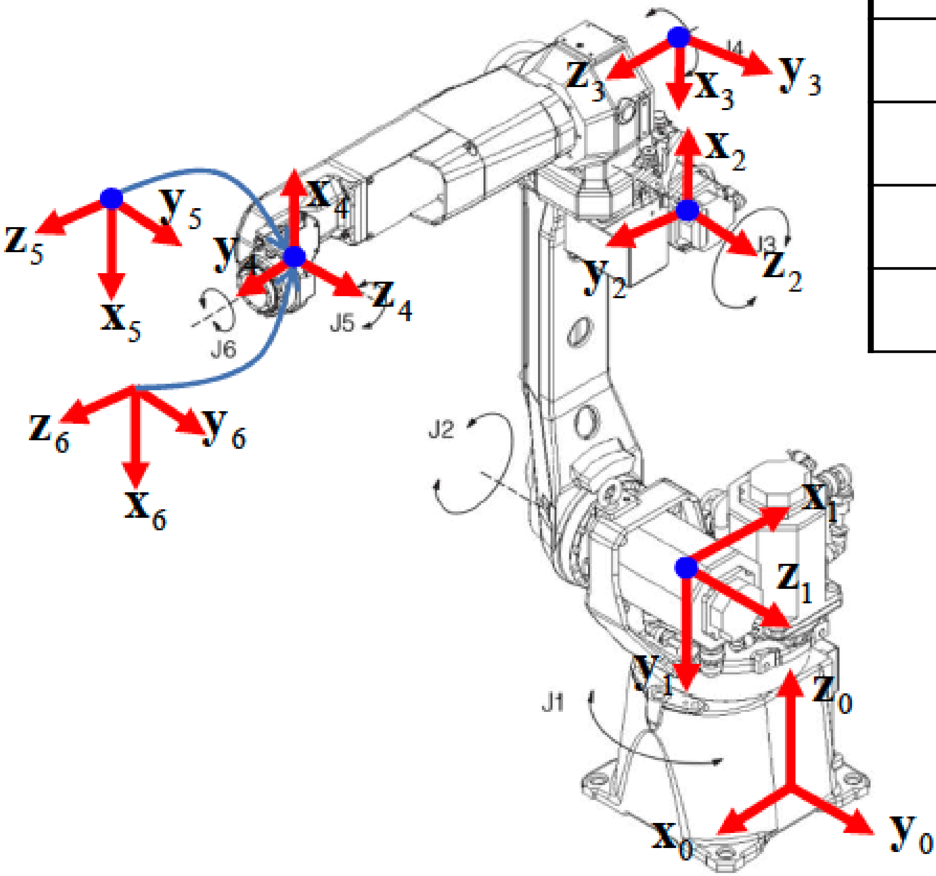
https://www.youtube.com/watch?v=Vrxaz0_X3Qs

Exemplo 10: Fanuc M-10iA





Exemplo 10: Fanuc M-10iA



Elo	a_i	α_i	d_i	θ_i
1	*-150*	90	*450*	$\theta_1 + 180$
2	*600*	0	0	θ_2+90
3	*-200*	90	*0*	θ_3+180
4	*0*	90	*640*	$\theta_4 + 180$
5	*0*	90	*0*	θ_5+180
6	*0*	0	*0*	θ_6

NOTAÇÃO

$$T_4^0 = A_1 \cdots A_4 = \begin{bmatrix} c_{12}\boxed{c_4} + s_{12}s_4 & -c_{12}s_4 + s_{12}c_4 & 0 & a_1c_1 + a_2c_{12} \\ s_{12}c_4 - c_{12}s_4 & -s_{12}s_4 - \boxed{c_{12}}c_4 & 0 & a_1\boxed{s_1} + a_2\boxed{s_{12}} \\ 0 & 0 & -1 & d_1 - d_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\boxed{c_i = \cos\theta_i \ \textbf{ex:}\ c_4 = \cos\theta_4}$$

$$\boxed{s_i = \operatorname{sen}\theta_i \ \textbf{ex:}\ s_1 = \operatorname{sen}\theta_1}$$

$$\boxed{c_{ij} = \cos(\theta_i + \theta_j)\ \textbf{ex:}\ c_{12} = \cos(\theta_{12}) = \cos(\theta_1 + \theta_2)}$$

$$\boxed{s_{ij} = \operatorname{sen}(\theta_i + \theta_j)\ \textbf{ex:}\ s_{12} = \operatorname{sen}(\theta_{12}) = \operatorname{sen}(\theta_1 + \theta_2)}$$

BIBLIOGRAFIA BÁSICA UTILIZADA

- SPONG M.W. Hutchinson, S., and Vidyasagar, M., "Robot Modeling and Control", John Wiley and Sons, 2006.
- Corke, Peter. Robotics, vision and control: fundamental algorithms in MATLAB. Vol. 73. Springer, 2011.
- Siciliano, Bruno, et al. *Robotics: modelling, planning and control*. Springer Science & Business Media, 2010.
- Jazar, Reza N. *Theory of applied robotics: kinematics, dynamics, and control*. Springer Science & Business Media, 2010.